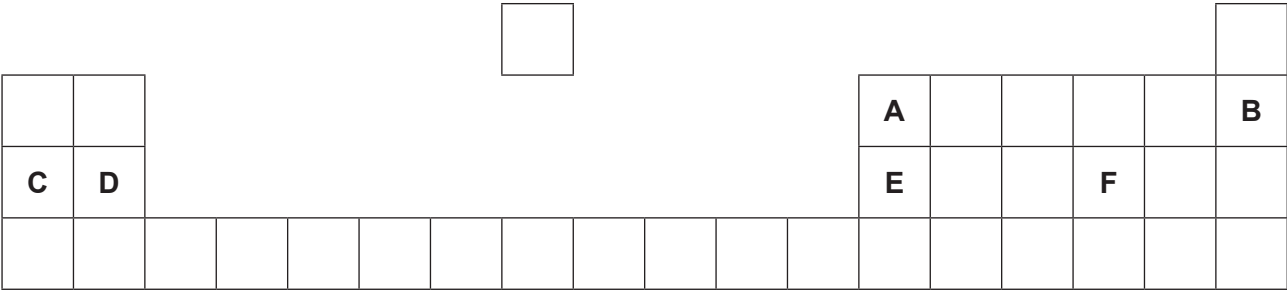


Answer **all** questions.

1. (a) The following diagram shows an outline of part of the Periodic Table.
The letters shown are NOT the chemical symbols of the elements.

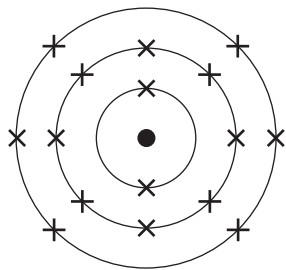


Choose **letters** from the diagram to complete the table below. [4]

	Letter
The element in Group 3 and Period 2	
The element which has 10 protons in its nucleus	
The element with the electronic structure 2,8,6	
The element which forms a 2+ ion	



(b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it.

[1]

(c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	³¹ X ₁₅		16	15
Y	³⁹ Y ₁₉	19		19
Z	⁴⁰ Z ₁₉	19	21	

(i) Complete the table.

[3]

(ii) Underline the term used to describe atoms **Y** and **Z**.

[1]

ions inert insoluble isotopes

